

POLITECNICO DI TORINO

Master's Degree in Electronics



Master's Degree Thesis

Loss analysis of a 4-level, flying capacitor, bridge-less, totem-pole PFC

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iTCM simulations

In order to prove that the the iTCM technique is in fact beneficial for the system in object, a simplified version of the PFC has been taken into account, all the major components are reported as is, however, not all parasitics of the components have been taken into account in order to speed up the simulations.

The GaN FETs model is the same as the one used in previous simulations, doing so it is possible to make a fair comparison between the two systems, concerning the devices losses, in particular, it is interesting to see the behavior of the GaNFET during all commutations, and how the switching losses will be affected.

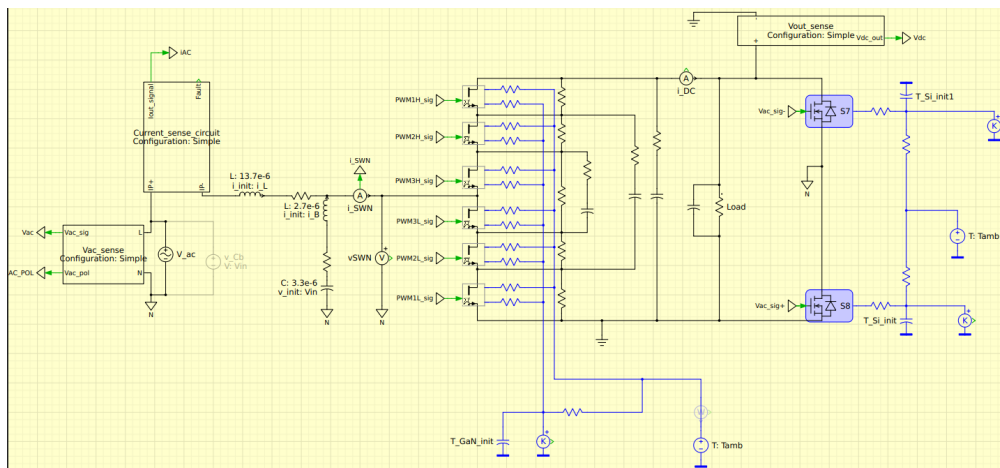


Figure 1.1: iTCM PFC model

The simulation is set-up in this way:

- A sinusoidal voltage reference is used to emulate the grid voltage behavior
- Input and output voltages are monitored in order to give a rudimental duty cycle estimation
- An artificial sinusoidal signal(with same frequency as the voltage reference) is used in order to generate the current reference

- A finite state machine monitors all the previous values and sends the commands to all the FETs in the correct sequence(including deadtimes)

1.1 FETs switching behaviour analysis

1.1.1 Low Side FETs

The most important feature of this system is the ability of soft switching the GaNFETs, which is proven by the following picture [1.2]

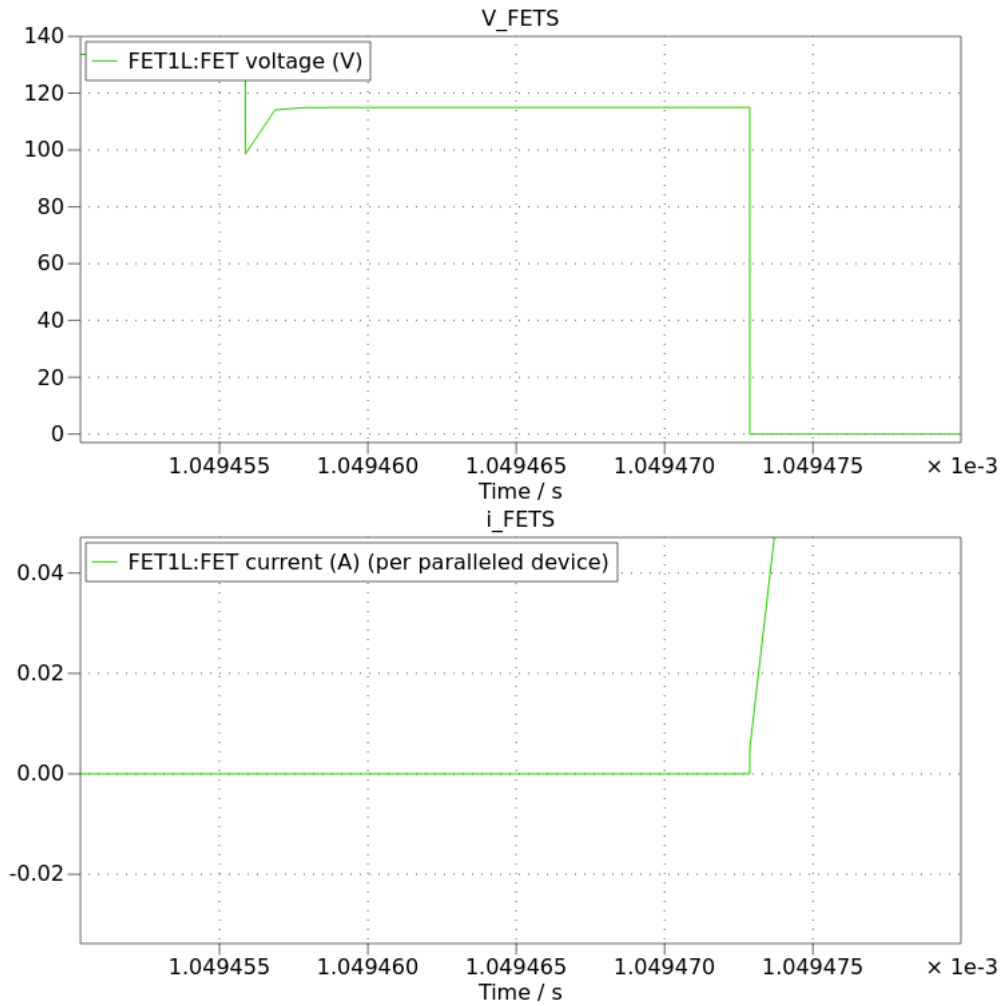


Figure 1.2: FET1L turn on behaviour

This trend is proven for all Low side FETs

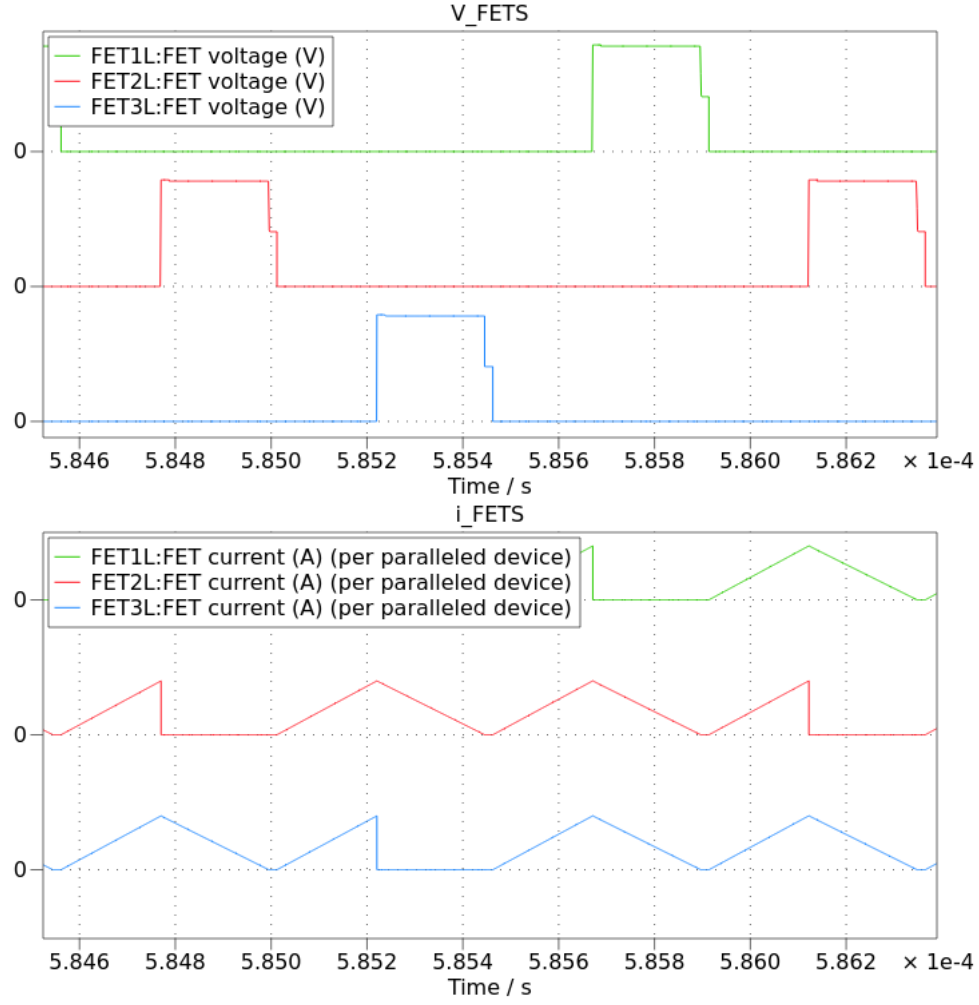
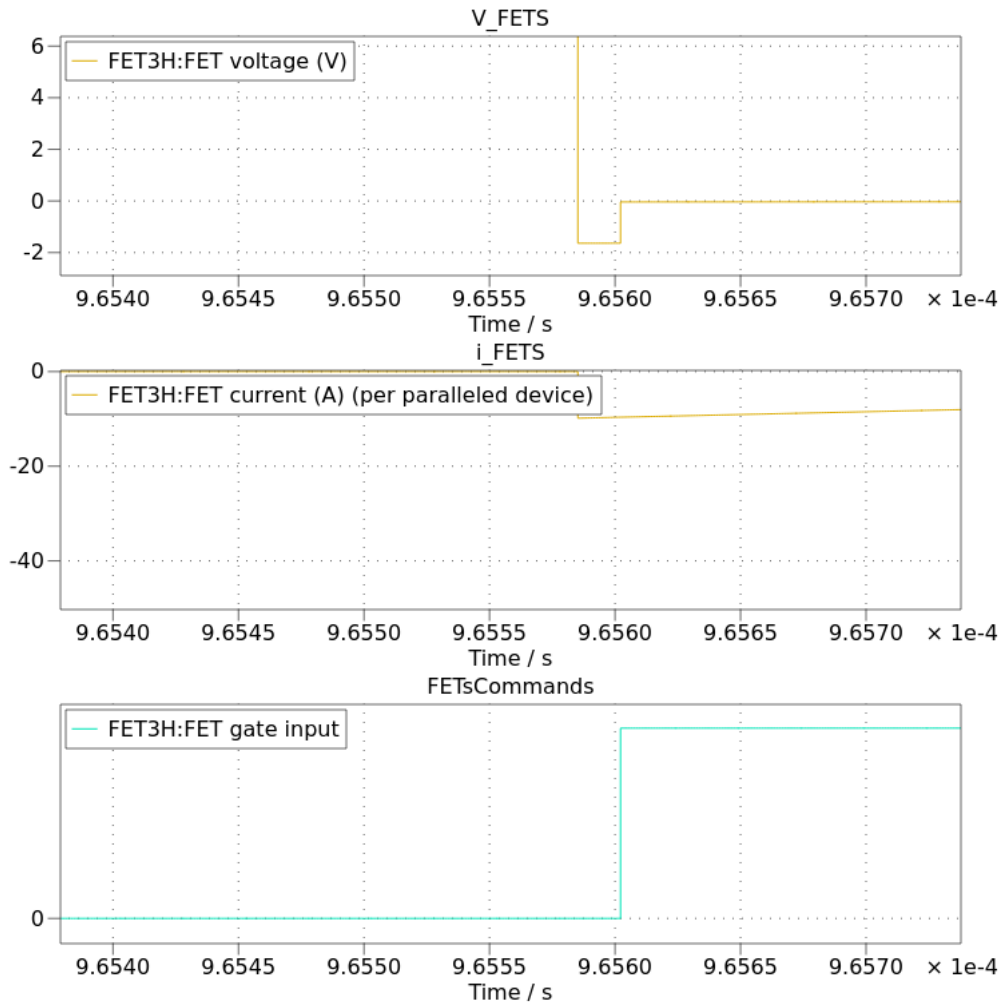


Figure 1.3: L FETs behaviour

This shows that the current reaches values of $\approx 0A$ before the device switch-on (abrupt change of device voltage), this allows to minimize the power dissipation inside the device.

1.1.2 High Side FETs

Regarding the high side FETs we can see that the device switch-off occurs $\approx 0A$ [1.5] while the switch-on occurs $\approx 0V$ due to the behavior of the virtual body diode of the device [1.4]

**Figure 1.4:** FET3H turn on behaviour

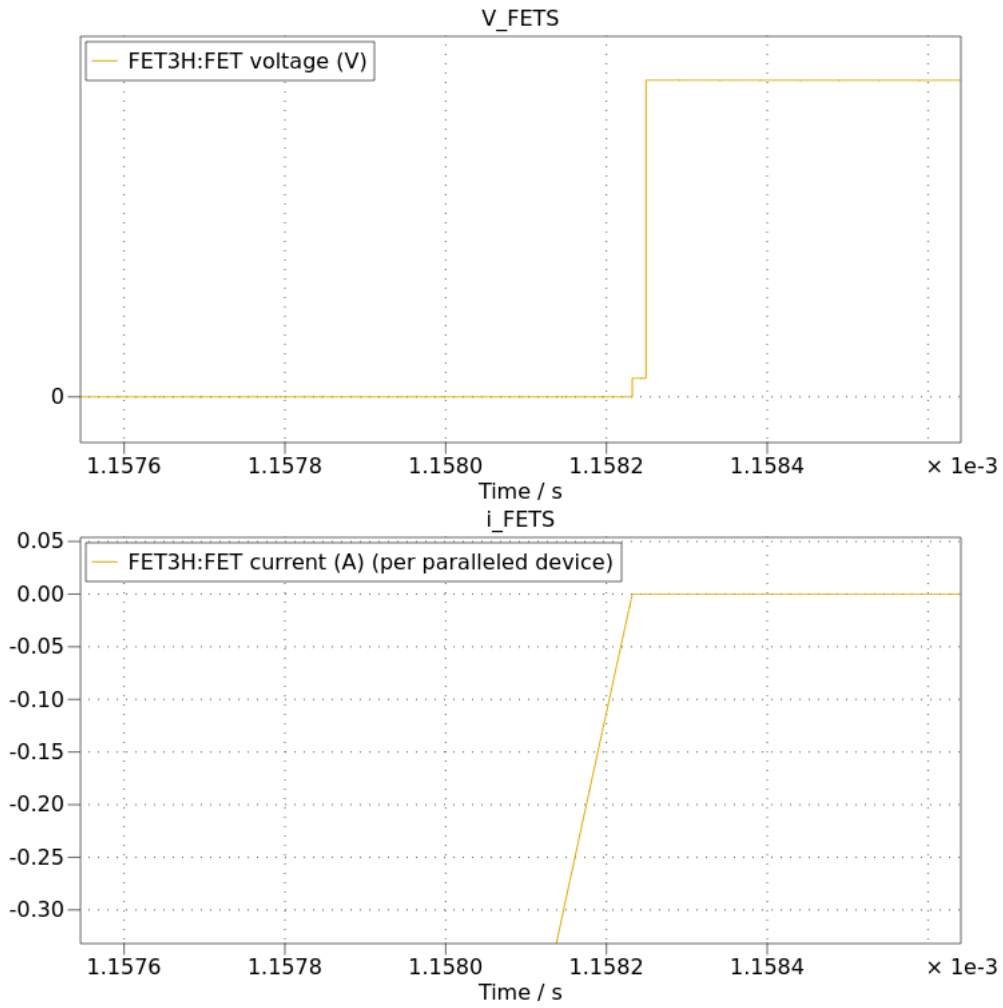


Figure 1.5: FET3H turn off behaviour

1.2 SWitching Node behavior

The behavior of the current in the switching node is shown below.

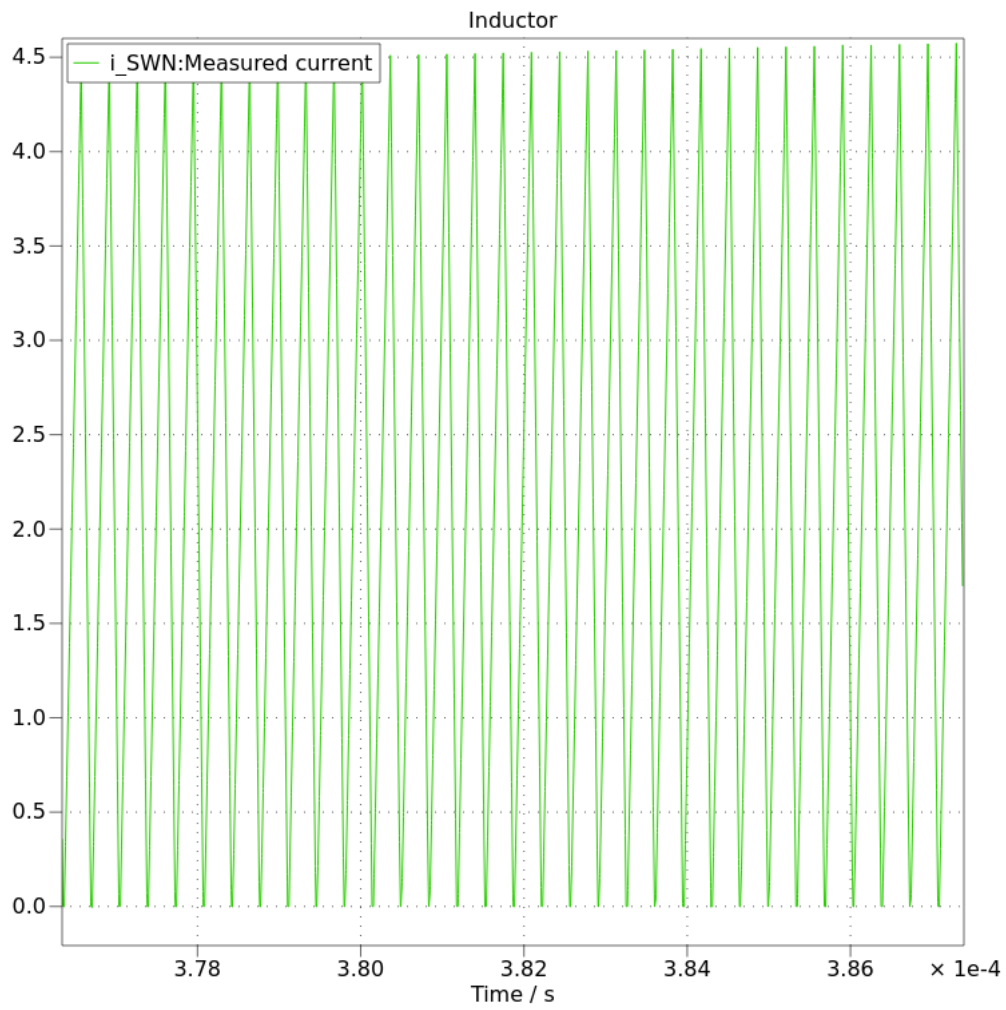


Figure 1.6: iSWN current detail

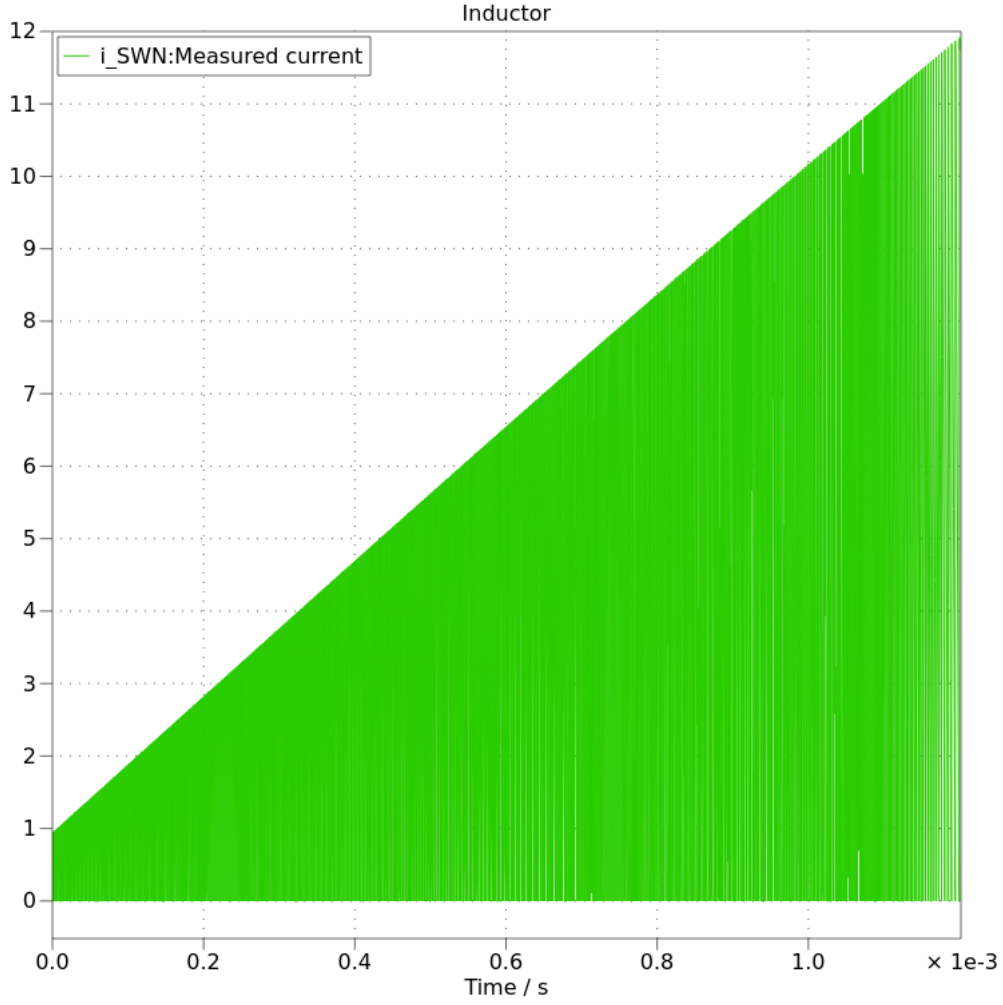


Figure 1.7: iSWN trend over time

it can be noted that for a short period of time the current in the switching node is a triangular current waveform, which allows the sytem to perform in Boundary Conduction Mode, which in the end allows us to perform the soft switching commutations, but at the same time we can see that the envelope of that current is following a sinusoidal path which is dictated by the controller(in this case is a sinusoidal generator).

At the same time, by looking at the current in the main and auxiliary inductor

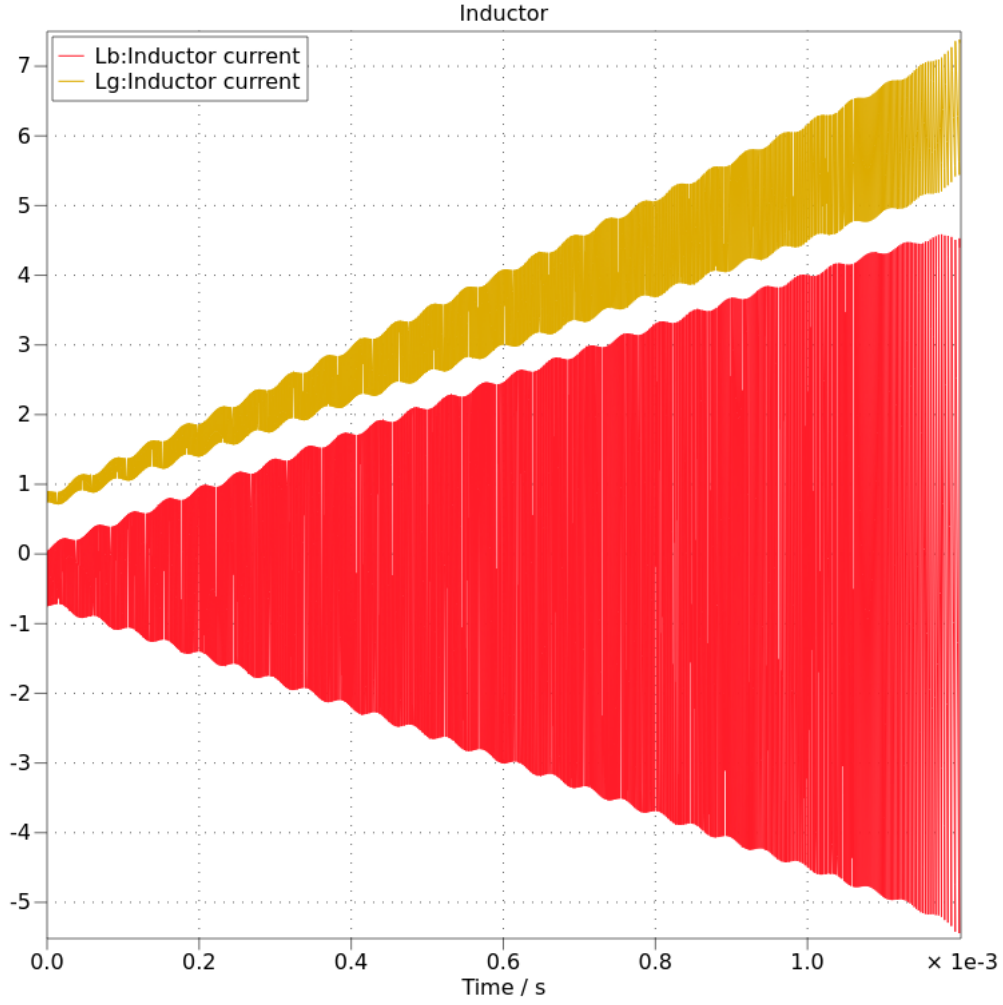


Figure 1.8: L_b and L_g currents

we can clearly see that the current in the main inductor L_g never reaches 0, during this segment of the grid cycle, which in the end allows us to choose an inductor core most suited for these working conditions, at the same time we can clearly see that a high ripple current is induced in the L_b inductor.

At last we can see the behavior of the voltage of the capacitor C_b which is following the grid voltage, with a ripple superimposed onto it (due to the L_b ripple)

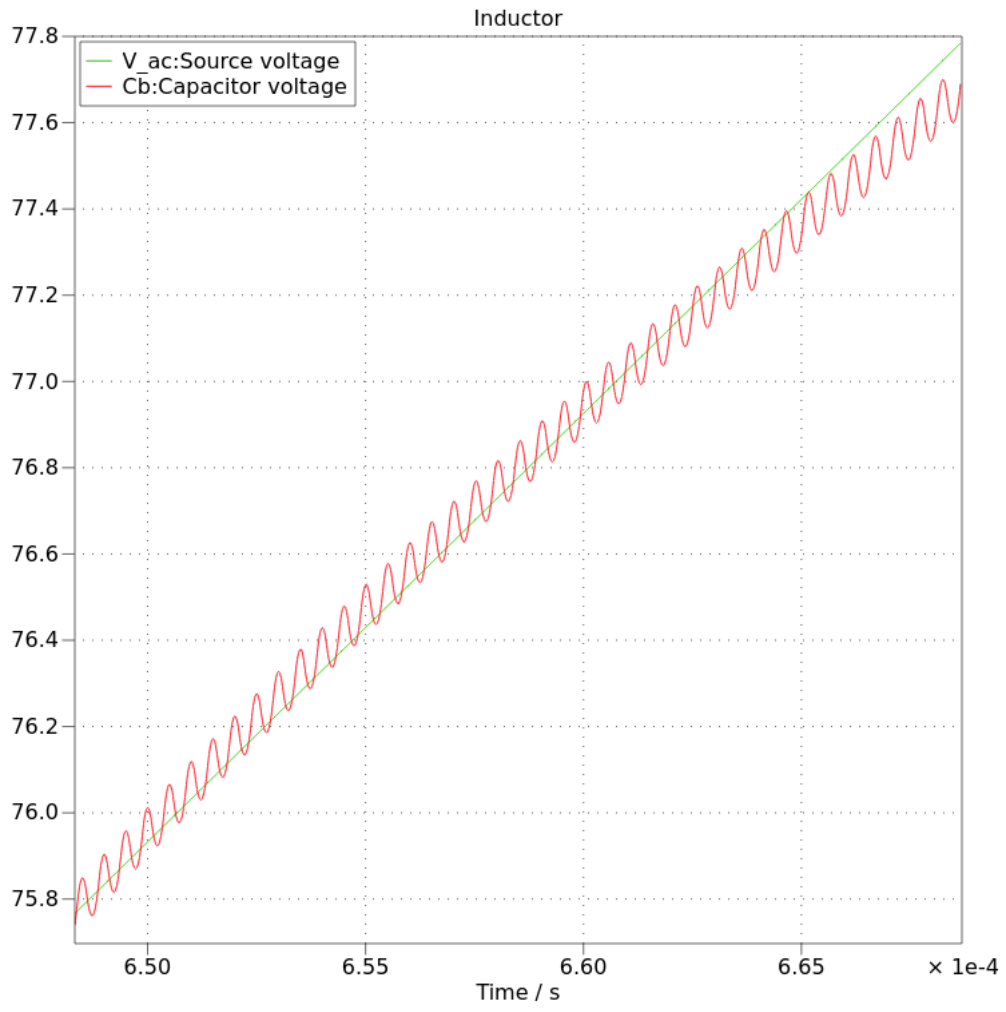


Figure 1.9: Voltage on C_b

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